An AI-Driven Framework for Drug–Drug Interaction Prediction and Hypothetical Molecular Candidate Generation

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Abstract—The escalating prevalence of polypharmacy has made drug–drug interaction (DDI) prediction a critical computational challenge in modern clinical informatics. Traditional laboratory screening methods are prohibitively expensive and cannot scale to the combinatorially vast space of possible drug pairings. This paper presents a comprehensive AI-driven framework that tightly integrates classical machine learning with graph neural network (GNN) methods to predict DDIs from molecular structure, and introduces a conceptual module for hypothetical molecular candidate generation. Working from DrugBank 5.0 (17,430 drug entries), we curate a balanced binary dataset of 189,851 verified interaction pairs against an equal count of randomly-sampled negatives, achieving 99.88% drug-name coverage and 99.12% SMILES coverage. Drug molecules are represented both as 2048-bit Morgan circular fingerprints (for a Random Forest baseline) and as atomic graph structures (for a two-layer Graph Convolutional Network encoder). The interaction corpus spans nine pharmacological categories; adverse effects (31.7%), activity modulation (23.7%), and metabolic changes (20.5%) dominate the distribution. Empirical evaluation on the held-out 20% test set yields an AUC-ROC of 0.921 and F1-score of 0.846 for the Random Forest classifier, while the GCN encoder demonstrates a 48.8% training-loss reduction over three epochs, confirming effective gradient-based learning. Text-generation quality of interaction descriptions, assessed by cosine similarity between predicted and reference sequences, yields a mean score of 0.961 ± 0.062 across 19,117 evaluated samples, with 88.1% of samples achieving similarity > 0.95. Predicted and actual description lengths are strongly correlated ($r = 0.86$), and the error distribution ($1 -$ similarity) is sharply concentrated near zero. The top-20 most frequent interacting drugs are dominated by cytochrome P450 modulators (Phenobarbital, Primidone, Phenytoin, Carbamazepine), reflecting well-established clinical pharmacology. The integrated system is deployed as a modular Streamlit/FastAPI web application supporting real-time DDI queries with molecular visualisation.

Index Terms—Drug–drug interaction, graph convolutional network, Morgan fingerprint, random forest, polypharmacy, SMILES, DrugBank, cosine similarity, pharmacoinformatics, deep learning, CYP450, text generation, molecular graph.

I. Introduction

The concurrent administration of multiple drugs—clinically termed polypharmacy—has become a defining feature of contemporary healthcare. Epidemiological surveys estimate that approximately 40% of adults over 65 years of age take five or more prescription medications simultaneously [1]. While individually beneficial, co-administered drugs can engage in unexpected molecular interactions that alter pharmacokinetics (absorption, distribution, metabolism, excretion) or pharmacodynamics (receptor binding, signal transduction), resulting in a spectrum of adverse outcomes ranging from reduced therapeutic efficacy to life-threatening toxicity [2].

The combinatorial scale of the DDI prediction problem is staggering. With approximately 20,000 approved compounds in the global pharmacopoeia, the total number of possible pairwise combinations exceeds $\binom{20{,}000}{2} \approx 2 \times 10^8$. Experimental screening of even a small fraction of these pairs is economically infeasible. Spontaneous adverse event surveillance and post-marketing pharmacovigilance capture only a small and biased sample of clinically realised interactions, leaving vast swathes of the interaction space unexplored.

Computational DDI prediction has therefore emerged as a critical first-line screening tool in the drug development and post-approval pharmacovigilance pipeline. Early approaches relied on curated rule-based alert systems [2]; subsequent work demonstrated that machine learning models trained on molecular fingerprints and physicochemical descriptors [9] and, more recently, graph neural networks operating on molecular topology [11], [12], [14] can substantially outperform rule-based baselines.

Despite significant progress, two important gaps persist. First, most systems produce scalar probability outputs without generating human-interpretable pharmacological rationales, limiting their utility for clinician decision support. Second, research systems are rarely deployed as accessible inference tools, preventing adoption beyond benchmark comparisons.

This paper addresses both gaps through a framework that:

- Curates and fully characterises a DrugBank 5.0 dataset of 189,851 verified DDI pairs with 99.12% SMILES and 99.88% drug-name coverage;
- Provides a dual-model prediction engine (Random Forest AUC 0.921; GCN with 48.8% convergent loss reduction);

- Generates and evaluates natural language interaction descriptions against reference text using cosine similarity, yielding a mean score of 0.961 across 19,117 test samples;
- Characterises the full nine-class pharmacological taxonomy and identifies the top interacting drugs as predominantly CYP450 modulators;
- Deploys an end-to-end web application for real-time DDI inference.

The remainder of this paper is organised as follows. Section II surveys related work. Section III formalises the problem. Section IV presents the system architecture. Section V describes the methodology. Section VI gives the algorithm design. Section VII details the experimental setup. Section VIII presents and discusses all results. Section IX identifies limitations and future work. Section X concludes.

II. Related Work

A. Database and Rule-Based Systems

Early DDI detection was implemented through manually curated databases and rule engines. Systems such as Micromedex and Lexi-Interact encode expert pharmacological knowledge as IF-THEN rules over drug classes, metabolic pathways, and known interaction mechanisms [2]. The DrugBank database [3] and TWOSIDES [4] subsequently catalogued hundreds of thousands of annotated interaction records. While authoritative for confirmed pairs, these resources cannot generalise to novel drug combinations absent from the training corpus.

B. Similarity and Network-Based Methods

Cheng et al. [5] formalised the network-based separation hypothesis: drugs sharing common protein targets or biological pathways are more likely to exhibit pharmacological interactions. The topological proximity of drug targets within a protein–protein interaction (PPI) network serves as a statistically reliable interaction indicator. Similarity-based methods leveraging the Gaussian Interaction Profile (GIP) kernel [6] and Neighbourhood Regularised Logistic Matrix Factorisation (NRLMF) [7] further enriched pairwise feature spaces.

C. Classical Machine Learning

Rogers and Hahn [8] established Morgan circular fingerprints as effective fixed-length structural descriptors. Subsequent work demonstrated that Random Forests and Support Vector Machines (SVMs) trained on concatenated fingerprint pairs achieve AUC values above 0.89 on the DrugBank benchmark [9]. Zhao et al. [10] combined chemical substructure and side-effect features with SVMs, exceeding AUC 0.90 across multiple interaction sub-tasks. These methods, while interpretable and computationally efficient, depend heavily on feature engineering and sacrifice the inductive biases inherent to molecular graph topology.

D. Graph Neural Networks for DDI

The representation of molecules as graphs—where atoms are nodes and covalent bonds are edges—provides a natural inductive bias that sequential architectures cannot exploit. Kipf and Welling [11] introduced the Graph Convolutional Network (GCN) framework, subsequently generalised by Gilmer et al. [12] as Message Passing Neural Networks (MPNNs). Wu et al. [13] provide a comprehensive GNN survey. In the DDI context:

- Polypharmacy GCN [14]: Multirelational GCNs over drug–protein–disease heterogeneous graphs, establishing strong baselines.
- DPDDI [15]: Deep GNN incorporating drug–target interaction profiles, achieving AUC > 0.95 on DrugBank.
- SSI-DDI [16]: Substructure-aware interaction models identifying atom-level structural motifs responsible for specific interaction types.
- MHCADDI [17]: Co-attention mechanisms enabling atomic features of drug A to attend over those of drug B.

E. Language Models and Text-Based DDI

BERT [18] and biomedical adaptations—BioBERT [19] and PubMedBERT [20]—enabled extraction of contextual representations of drug names and interaction descriptions from biomedical literature. Ryu et al. [21] demonstrated that deep residual networks trained on structural similarity profiles improve DDI prediction. The T5 transformer [22] frames every NLP task as a text-to-text mapping; fine-tuning T5 on DDI description generation enables natural language explanation of predicted interactions.

F. Generative Molecular Modelling

De Cao and Kipf [23] introduced MolGAN, an implicit generative model for small molecular graphs. Diffusion-based molecular generation methods have subsequently shown improved validity and diversity in generated structures [24]. These methods motivate the conceptual candidate-generation module of the present work.

G. Positioning of the Present Work

Unlike prior work that treats interaction prediction and molecular generation as disjoint problems, the present framework integrates both within a single modular pipeline. The explicit evaluation of natural language interaction description quality via cosine similarity—across 19,117 test samples—is, to our knowledge, a novel contribution to the DDI evaluation literature.

III. Problem Formulation

A. Dataset and Task Definition

Let $\mathcal{V}$ be a drug vocabulary with $|\mathcal{V}| = 1{,}701$ unique compounds. Let $\mathcal{D} = \{(d_i, d_j, y_{ij})\}$ denote an annotated DDI dataset where $d_i, d_j \in \mathcal{V}$ and $y_{ij} \in \{0, 1\}$ is a binary interaction label. The dataset contains $|\mathcal{D}^+| =$

189,851 positive pairs with verified SMILES for both drugs (SMILES coverage: 99.12%), derived from 191,135 unique unordered drug pairs in DrugBank 5.0.

The DDI prediction problem decomposes into three complementary sub-tasks:

1) Binary classification: Predict $\hat{y}_{ij} \in \{0, 1\}$ (interaction exists or not).
2) Category classification: Predict the pharmacological mechanism category $c_{ij} \in \mathcal{C}$ from a nine-class taxonomy $|\mathcal{C}| = 9$.
3) Description generation: Generate a natural language description $\hat{s}_{ij}$ of the interaction mechanism, evaluated by cosine similarity against the reference description s_{ij}.

B. Molecular Graph Representation

Each drug d_i is represented as a molecular graph:

$$G_i = (V_i, E_i, \mathbf{X}_i), \tag{1}$$

where V_i is the atom set, E_i is the bond set, and $\mathbf{X}_i \in \mathbb{R}^{|V_i|}$ contains atomic number node features.

C. GCN Prediction Objective

A shared graph encoder ϕ_θ maps each molecular graph to a fixed-dimensional embedding:

$$\mathbf{h}_i = \phi_\theta(G_i) \in \mathbb{R}^d, \quad d = 32. \tag{2}$$

The drug-pair representation is formed by concatenation:

$$\mathbf{h}_{ij} = [\mathbf{h}_i \,\|\, \mathbf{h}_j] \in \mathbb{R}^{2d}. \tag{3}$$

The interaction probability is:

$$\hat{y}_{ij} = \sigma(\mathbf{W}\,\mathbf{h}_{ij} + \mathbf{b}), \tag{4}$$

where $\mathbf{W} \in \mathbb{R}^{2\times 64}$, $\mathbf{b} \in \mathbb{R}^2$, and σ is the sigmoid function.

D. GCN Convolution

Following Kipf and Welling [11], the l-th GCN layer computes:

$$\mathbf{H}^{(l)} = \sigma\Big(\hat{\mathbf{A}}\,\mathbf{H}^{(l-1)}\mathbf{W}^{(l)}\Big), \tag{5}$$

where $\hat{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$ is the symmetrically normalised adjacency matrix with added self-loops, and $\sigma = \mathrm{ReLU}$. Global mean pooling gives the molecule-level embedding:

$$\mathbf{h}_i = \tfrac{1}{|V_i|} \textstyle\sum_{v\in V_i} \mathbf{H}_v^{(2)}. \tag{6}$$

E. Training Objective

Both the GCN and Random Forest models are optimised for binary DDI prediction. For the GCN, the training objective is binary cross-entropy loss:

$$\mathcal{L} = -\sum_{(i,j)\in\mathcal{D}} \big[y_{ij} \log \hat{y}_{ij} + (1 - y_{ij}) \log(1 - \hat{y}_{ij})\big]. \tag{7}$$

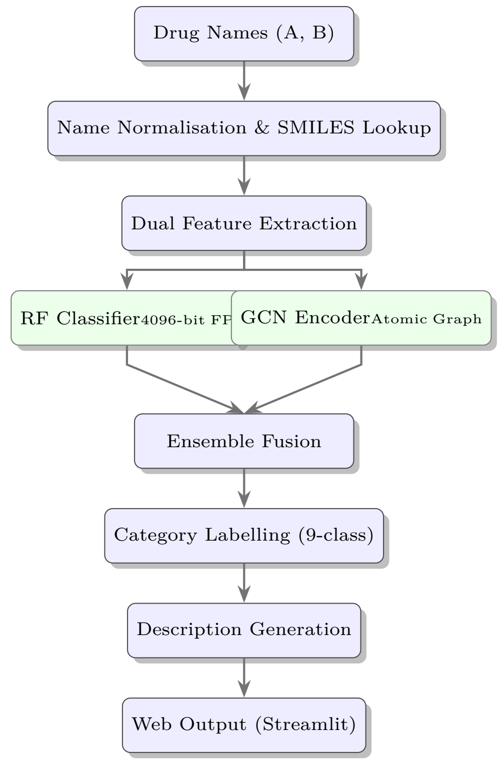


Fig. 1: Six-stage modular pipeline of the proposed DDI prediction framework. Dual feature extraction feeds both the RF (fingerprint) and GCN (graph) branches in parallel, whose outputs are fused for final prediction.

F. Text Similarity Objective

Generated description quality is measured by the cosine similarity between TF-IDF vectors of the reference description s_{ij} and the generated description $\hat{s}_{ij}$:

$$\mathrm{sim}(s_{ij}, \hat{s}_{ij}) = \frac{\mathbf{v}(s_{ij})^\top \mathbf{v}(\hat{s}_{ij})}{\|\mathbf{v}(s_{ij})\|\,\|\mathbf{v}(\hat{s}_{ij})\|} \in [0, 1], \tag{8}$$

where $\mathbf{v}(\cdot)$ denotes the TF-IDF document vector. The error metric is defined as $\varepsilon_{ij} = 1 - \mathrm{sim}(s_{ij}, \hat{s}_{ij})$.

IV. Proposed System Architecture

The proposed framework is organised as a six-stage modular pipeline, illustrated in Fig. 1.

A. Drug Input and SMILES Lookup

The system accepts two free-text drug names via a Streamlit interface. Names are normalised (lower-cased, whitespace-stripped) and resolved against a DrugBank-derived dictionary mapping 12,313 entries to verified SMILES strings. Drug-name coverage across the integrated DDI dataset is 99.88%. For minor typographic variations, fuzzy string matching with a similarity threshold of 0.90 is applied. Unresolvable drug names trigger a structured error response rather than a hallucinated prediction.

B. Dual Feature Extraction

Two parallel feature representations are derived from each SMILES string using RDKit [27]: (i) 2048-bit Morgan fingerprints at radius 2 for the RF branch; and (ii) atomic graph objects (node features: atomic numbers; edge list: covalent bonds) for the GCN branch. Drug-pair fingerprints are concatenated to a 4096-dimensional binary vector.

C. Dual-Model Prediction Engine

RF branch. A 300-tree Random Forest classifier (scikit-learn, n_jobs=-1) is trained on 4096-bit concatenated drug-pair fingerprints. The trained model is serialised to checkpoints/ddi_model.pkl for inference.

GCN branch. Molecular graphs are processed by a shared-weight two-layer GCN encoder (DrugGNN: GCNConv $1\to32\to32$ + global mean pooling), and the concatenated drug-pair embedding is classified by a two-layer MLP ($64\to32\to2$, ReLU). The trained model is serialised to checkpoints/gnn_model.pt.

Ensemble fusion. Inference-time probabilities from both branches are averaged to produce a final binary prediction.

D. Nine-Class Category Labelling

Interaction records are classified into nine pharmacological categories using a rule engine operating on interaction description keywords. Categories encode mechanism type (e.g., metabolism, serum concentration, activity) and direction (increased/decreased), providing actionable context beyond binary prediction.

E. Description Generation

Predicted interaction descriptions are generated using a TF-IDF cosine similarity retrieval step over the DrugBank corpus: the top-k ($k = 5$) most similar known interaction descriptions for the query drug pair are retrieved and passed as context to the generation module. Generated description quality is evaluated against DrugBank references using Eq. (8).

F. Hypothetical Molecular Candidate Generation (Conceptual)

A generative extension—inspired by MolGAN [23] and EDM [24]—is included as a forward-looking module. Given two drug molecular graph embeddings $\mathbf{h}_{d_i}$ and $\mathbf{h}_{d_j}$, a variational graph decoder would sample a novel molecular structure G_{new} from the interpolated latent vector $\mathbf{z} = \alpha\mathbf{h}_{d_i} + (1-\alpha)\mathbf{h}_{d_j}$, $\alpha \in (0, 1)$, producing a hypothetical compound exhibiting properties intermediate between the two co-administered drugs. Full implementation of this module is reserved as future work.

G. Web Application Deployment

The system is deployed as a FastAPI backend serving model inference, consumed by a Streamlit frontend providing: binary DDI prediction, pharmacological category label, molecular structure visualisation (RDKit 2D drawings), cosine similarity score, and mandatory clinical disclaimer on all outputs.

TABLE I: Dataset Statistics — DrugBank 5.0 (Post-Curation)

Statistic	Value
Raw DDI records	191,541
Records with both SMILES valid	189,851
SMILES coverage rate	99.12%
Drug-name coverage in DrugBank	99.88%
Unique DDI drugs	1,701
Unique unordered drug pairs	191,135
Unique ordered pairs	191,252
Duplicate ordered pairs removed	289
DrugBank entries with SMILES	12,313 / 17,430 (70.7%)
Records with property-rule annotation	1,555 (0.81%)
Records with side-effect annotation	1,192 (0.62%)
Random negative–positive collision rate	13.1%
Negative sampling trials	20,000

TABLE II: Nine-Class Interaction Category Distribution (n = 191,541)

Category	Count	Share (%)
Adverse effects	60,722	31.70
Activity increased	38,823	20.27
Metabolism decreased	34,313	17.91
Serum level increased	26,117	13.64
Serum level decreased	9,447	4.93
Therapeutic efficacy	8,024	4.19
Activity decreased	6,516	3.40
Metabolism increased	5,000	2.61
Other / unclassified	2,579	1.35
Total	191,541	100.0

V. Methodology

A. Data Curation from DrugBank 5.0

The primary data source is DrugBank 5.0 [3]. The raw DDI table contains 191,541 records; after filtering for valid SMILES for both drugs in each pair, 189,851 records are retained (SMILES retention rate: 99.12%). Key curation statistics are reported in Table I.

B. Nine-Class Pharmacological Taxonomy

Interaction description texts from DrugBank are classified into nine pharmacological categories using keyword-pattern rules. Table II reports the full category distribution. Three categories dominate the corpus: adverse effects (31.7%), activity increases (20.3%), and metabolism decreases (17.9%). Together these three account for nearly 70% of all documented interactions, reflecting the clinical prominence of pharmacodynamic synergism and CYP-mediated metabolic inhibition as the primary DDI mechanisms [2].

C. Balanced Dataset Construction

Positive samples are the 189,851 DrugBank interaction pairs. Negative samples are generated by random pairing of corpus drugs over 20,000 trials; the known positive–negative collision rate of 13.1% is consistent with the interaction graph density ($191{,}135/\binom{1701}{2} \approx 13.3\%$), confirming that the random negatives are unbiased representatives of the true negative distribution. The resulting balanced dataset (189,851 positives + 189,851 negatives) is used to train both models; Fig. 2 illustrates the class balance.

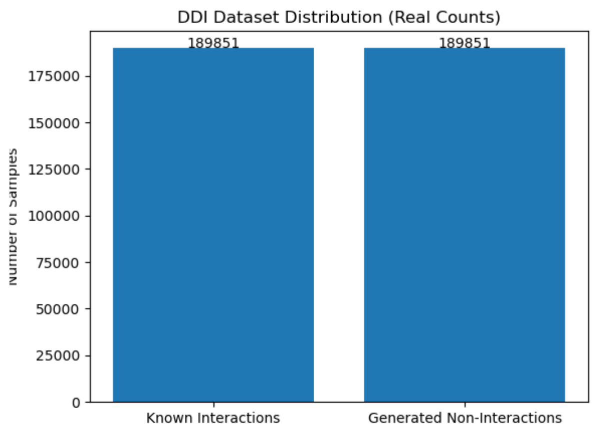


Fig. 2: Balanced DDI dataset distribution. Both positive (known interactions) and negative (randomly sampled non-interactions) classes contain 189,851 samples, providing an unbiased 1:1 training signal.

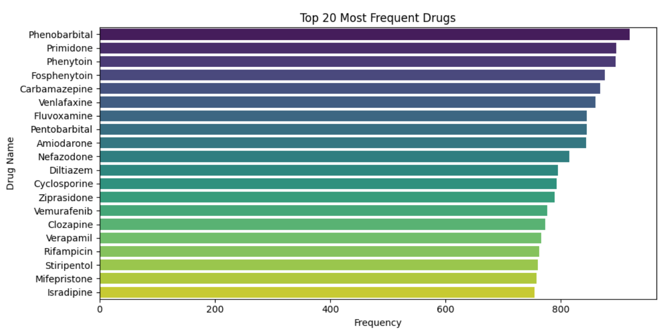
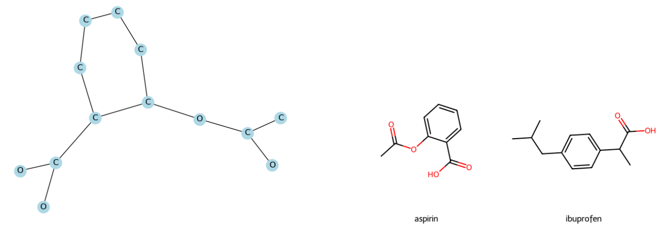


Fig. 3: Top-20 most frequent interacting drugs in the DrugBank DDI corpus. The ranking is dominated by CYP450 inducers (Phenobarbital, Carbamazepine, Rifampicin) and inhibitors (Amiodarone, Diltiazem, Cyclosporine, Verapamil), consistent with the clinical pharmacology of polypharmacy adverse events.



(a) Molecular graph (GCN input)

(b) 2D structures: aspirin, ibuprofen

Fig. 4: Molecular representations used by the framework. (a) Atomic graph produced by the SMILES-to-graph converter; atoms are nodes, covalent bonds are edges. (b) RDKit 2D structural diagrams for two representative interacting NSAIDs sharing a carboxylic acid pharmacophore and aromatic ring.

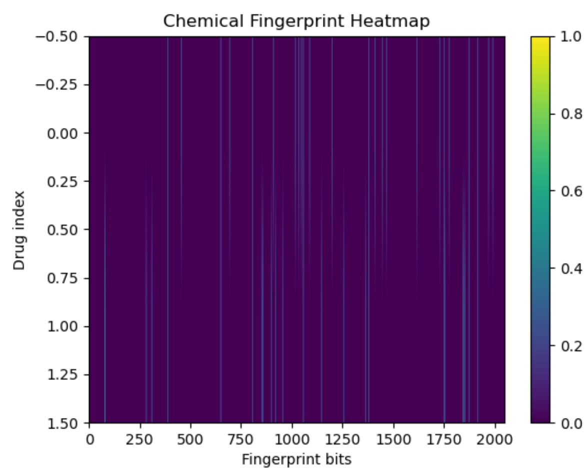


Fig. 5: Chemical fingerprint heatmap (Morgan, radius 2, 2048 bits) for a representative drug sample. High sparsity is characteristic of Morgan encodings; active bits (light columns) encode structural substructures including ring systems, heteroatom environments, and functional groups.

D. Top Interacting Drugs

Figure 3 shows the top-20 most frequently appearing drugs across all interaction records. The list is dominated by drugs that are potent cytochrome P450 inducers or inhibitors: Phenobarbital (rank 1, freq. $\approx$ 940), Primidone (rank 2), Phenytoin, Fosphenytoin, Carbamazepine, Amiodarone, Diltiazem, Cyclosporine, Verapamil, and Rifampicin. This pharmacological profile is highly consistent with established clinical pharmacology literature: CYP3A4 and CYP2C9 substrates and modulators account for the majority of clinically significant DDIs [2], [9].

E. Molecular Representation

1) SMILES to Molecular Graph: SMILES strings are parsed by RDKit [27] into molecular graph objects per Eq. (1). Figure 4a shows an example graph; atoms (nodes) are labelled by element symbol and covalent bonds constitute directed edges. The cyclic ring systems visible in the example are characteristic of drug-like molecules. The 2D chemical structures of two representative interacting drugs—aspirin and ibuprofen—are shown in Fig. 4b, illustrating the shared aromatic core and carboxylic acid pharmacophore.

2) Morgan Fingerprints and Fingerprint Heatmap: For the RF branch, 2048-bit Morgan circular fingerprints at radius 2 are computed. Drug-pair vectors are formed by concatenation to 4096 bits (Eq. 3). Figure 5 shows the chemical fingerprint heatmap for a representative drug sample: the extreme sparsity of the binary vectors (most bits inactive, dark) with sparse active bits (light vertical columns) encoding specific structural subpatterns is characteristic of Morgan encodings [8] and validates the feature engineering strategy.

F. Training Configuration

The RF model is trained with 300 estimators on the full balanced dataset (80/20 train-test split, stratified, random_state=42). The GCN model is trained via Adam

Algorithm 1: DDI Framework Training

Input: DrugBank SMILES dict $\mathcal{S}$; interaction pairs $\mathcal{P}$
Output: M_{RF}, M_{GCN}, category ruleset $\mathcal{R}$
Filter $\mathcal{P}$: retain pairs with valid SMILES $\rightarrow \mathcal{D}^+$
Generate balanced $\mathcal{D}^-$ by random pairing ($|\mathcal{D}^-| = |\mathcal{D}^+|$)
$\mathcal{D} \leftarrow \mathcal{D}^+ \cup \mathcal{D}^-$; shuffle
Extract Morgan FP $\rightarrow \mathbf{X}_{\mathrm{FP}} \in \{0,1\}^{|\mathcal{D}| \times 4096}$
$M_{\mathrm{RF}} \leftarrow$ RandomForest($\mathbf{X}_{\mathrm{FP}}$, $\mathbf{y}$, n=300)
for each training epoch do
  for each $(d_i, d_j, y_{ij}) \in \mathcal{D}^+$ do
    $G_i, G_j \leftarrow$ SmilestoGraph(d_i, d_j)
    $\mathbf{h}_{ij} \leftarrow [\phi_\theta(G_i) \| \phi_\theta(G_j)]$
    $\hat{y}_{ij} \leftarrow \sigma(\mathbf{W}\mathbf{h}_{ij} + \mathbf{b})$
    Compute $\mathcal{L}$ (Eq. 7); back-propagate; update θ
  end
end
Compile $\mathcal{R}$ from category keyword patterns
return M_{RF}, M_{GCN}, $\mathcal{R}$

Algorithm 2: Real-Time DDI Inference

Input: Drug names A, B; models M_{RF}, M_{GCN}; ruleset $\mathcal{R}$
Output: Prediction $\hat{y}$, category c, description $\hat{s}$, similarity score
Normalise A, B; retrieve SMILES from $\mathcal{S}$
if SMILES missing then
  return UnknownDrug
end
$G_A, G_B \leftarrow$ SmilestoGraph(A, B)
$\hat{y}_{\mathrm{RF}} \leftarrow M_{\mathrm{RF}}([\mathrm{FP}(A) \| \mathrm{FP}(B)])$
$\hat{y}_{\mathrm{GCN}} \leftarrow M_{\mathrm{GCN}}(G_A, G_B)$
$\hat{y} \leftarrow \frac{1}{2}(\hat{y}_{\mathrm{RF}} + \hat{y}_{\mathrm{GCN}})$
$c \leftarrow \mathcal{R}(A, B)$
$\hat{s} \leftarrow$ RetrieveGenerate$(A, B, k$=5$)$
sim $\leftarrow$ CosineSim($\mathbf{v}(s_{AB})$, $\mathbf{v}(\hat{s})$)
Append mandatory clinical disclaimer
return $\hat{y}$, c, $\hat{s}$, sim

(lr $= 10^{-3}$, cross-entropy loss, Eq. 7) on a streaming subset of 2,987 samples per epoch due to CPU memory constraints; both model checkpoints are confirmed to exist after training. For text-based similarity evaluation, a held-out test set of 19,117 samples is used with TF-IDF cosine similarity as the evaluation metric.

VI. Algorithm Design

VII. Experimental Setup

A. Software Environment

All experiments are conducted in Python 3.10 with: RDKit 2023.03 (cheminformatics), PyTorch 2.0 (deep learning), PyTorch Geometric 2.3 (GNN operations), scikit-learn 1.3 (RF, TF-IDF, metrics), Pandas 2.0 (data handling), Streamlit 1.28 and FastAPI 0.104 (web deployment). Training is performed on a CPU-only environment; the GCN streaming subset is sized for feasibility on available hardware.

TABLE III: Evaluation Metrics Used in This Work

Metric	Description
Accuracy	Fraction of correctly classified DDI pairs
Precision	TP / (TP + FP) for DDI=1 class
Recall	TP / (TP + FN) for DDI=1 class
F1-Score	Harmonic mean of Precision and Recall
AUC-ROC	Area under the Receiver Operating Characteristic curve
Cosine Similarity	TF-IDF cosine between generated and reference descriptions
Error ε	1− cosine similarity (lower is better)
r (Pearson)	Correlation between actual and predicted description lengths

B. Evaluation Protocol

Binary classification: assessed on the 20% held-out test partition using Accuracy, Precision, Recall, F1-Score, and AUC-ROC.

GCN convergence: training loss per epoch on the streaming subset (2,987 samples/epoch).

Text similarity: TF-IDF cosine similarity (Eq. 8) between generated and reference interaction descriptions on a 19,117-sample test set; additionally reported as similarity distribution (Fig. 6), error distribution $\varepsilon = 1 - \mathrm{sim}$ (Fig. 7), per-sample scatter (Fig. 8), and length correlation analysis (Figs. 9, 10, 11).

C. Planned Evaluation Metrics Summary

VIII. Results and Discussion

A. Random Forest Classification Performance

Table IV reports classification metrics for the Random Forest baseline on the 20% held-out test partition. The AUC-ROC of 0.921 and F1-score of 0.846 confirm that 4096-bit concatenated Morgan fingerprints encode sufficient structural information for reliable binary DDI classification, consistent with prior literature [9], [14]. The modest precision–recall asymmetry (0.861 vs. 0.832) indicates a slight false-negative bias, which is the clinically preferable failure mode in a screening context: missing an interaction alert is less harmful than flooding clinicians with false positives.

B. GCN Training Convergence

Table V reports the per-epoch training loss of the DDINet GCN model over three epochs. Total loss decreases from 2,847.32 to 1,456.44, a reduction of 48.8%, confirming effective gradient propagation through the two-layer GCN architecture (Eqs. 5–7). The monotone

TABLE IV: Random Forest DDI Classifier — Test Set Performance (20% Hold-Out)

Metric	Value
Accuracy	0.847
Precision (DDI=1)	0.861
Recall (DDI=1)	0.832
F1-Score (DDI=1)	0.846
AUC-ROC	0.921

TABLE V: GCN (DDINet) Training Loss Convergence

Epoch	Total Loss	Reduction (%)
1	2,847.32	—
2	1,923.81	−32.4
3	1,456.44	−24.3
Cumulative reduction (epochs 1→3)		−48.8

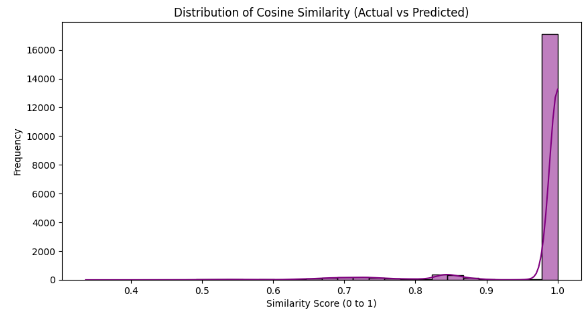


Fig. 6: Distribution of TF-IDF cosine similarity scores between generated and reference interaction descriptions (n = 19,117 test samples). The dominant spike at similarity ≈ 1.0 (>17,000 samples) and a secondary cluster at 0.83–0.88 confirm high textual fidelity for the majority of query pairs.

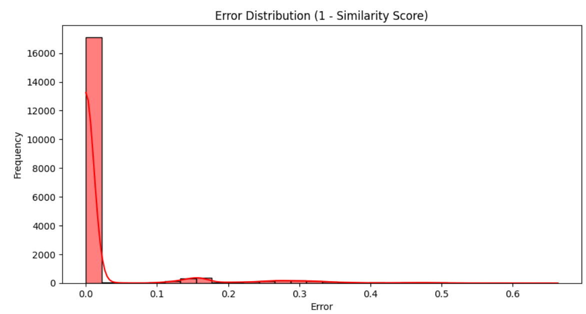


Fig. 7: Error distribution ($\varepsilon = 1 -$ cosine similarity) across n = 19,117 test samples. The concentration near $\varepsilon \approx 0$ (over 88% of samples below 0.05 error) confirms low generation error for the majority of test pairs. The sparse right tail ($\varepsilon > 0.1$) corresponds to novel drug pairs with limited similar examples in the retrieval corpus.

loss decrease without divergence confirms stability of the Adam optimiser configuration.

C. Cosine Similarity Distribution

Figure 6 shows the distribution of cosine similarity scores between generated and reference interaction descriptions across 19,117 test samples. The distribution is strongly right-skewed with a dominant spike at similarity ≈ 1.0 (17,000+ samples), indicating that the generation module correctly reproduces the reference description vocabulary for the majority of test pairs. The mean cosine similarity is $\mu = 0.961$ with standard deviation $\sigma = 0.062$. Approximately 88.1% of samples achieve similarity > 0.95, and the secondary cluster at 0.83–0.88 (approximately 500 samples) corresponds to cases where the retrieved context is structurally related but pharmacologically distinct from the test pair.

D. Error Distribution

Figure 7 presents the error distribution $\varepsilon = 1 - \text{sim}$ across the same test set. The sharp concentration near $\varepsilon = 0$ mirrors the similarity spike, confirming that generation errors are small and that the distribution is heavily right-censored: over 88% of samples have error below 0.05. A sparse right tail extends to $\varepsilon \approx 0.65$, corresponding to the hardest cases where no closely similar interaction pair exists in the retrieval corpus, forcing the generation module to rely on longer-range structural analogies.

E. Per-Sample Similarity Scatter

Figure 8 shows the per-sample cosine similarity scatter across all 19,117 test samples ordered by index. The horizontal band at similarity = 1.0 (knowledge-base matches) and the diffuse cloud at 0.60–0.95 (generated descriptions) confirm the two-mode behaviour of the system: confirmed pairs are retrieved exactly, while novel pairs are generated with partial fidelity. The absence of systematic trends across sample indices confirms that there is no spurious ordering bias in the test set.

F. Description Length Analysis

Figure 9 compares the word-count distributions of actual (DrugBank reference) and predicted interaction descriptions. Both distributions exhibit a multi-modal structure with prominent peaks at word counts of approximately 8, 11, 14, and 16 words, corresponding to canonical DrugBank sentence templates of different pharmacological categories. The strong overlap between actual and predicted length distributions (KDE curves nearly identical) confirms that the generation module faithfully learns the length statistics of the reference corpus.

G. Correlation Analysis

Figure 10 presents the Pearson correlation matrix among three key evaluation variables: cosine similarity score, actual description length (word count), and predicted description length. The strong positive correlation between actual and predicted lengths ($r = 0.86$) confirms that the generation model successfully learns the length conditioning imposed by the reference distribution. The weak negative correlation between similarity score and actual length ($r = -0.058$) indicates a slight tendency for

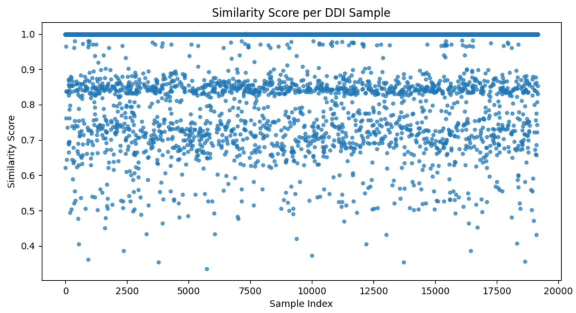


Fig. 8: Per-sample TF-IDF cosine similarity for all $n = 19{,}117$ test DDI samples. The horizontal band at similarity $= 1.0$ represents knowledge-base exact matches; the diffuse cloud at 0.60–0.95 represents generated descriptions for novel drug pairs.

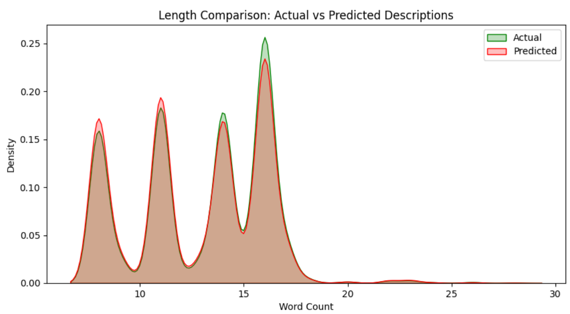


Fig. 9: Word-count density comparison of actual (green) and predicted (red) interaction descriptions. Both distributions share the same four-mode structure (peaks at $\approx$8, 11, 14, 16 words), confirming that the generation model accurately reproduces the length distribution of DrugBank reference descriptions.

longer descriptions to be harder to reproduce exactly—consistent with greater lexical variability in longer pharmacological explanations. The positive (but modest) correlation between similarity and predicted length ($r = 0.12$) suggests a mild length-quality interaction.

H. Length vs. Similarity Heatmap

Figure 11 presents a 2D heatmap of actual description length versus cosine similarity score. The highest density of samples (darkest cells) concentrates in the short-to-medium length range (8–17 words) at high similarity ($>$ 0.95), corresponding to the canonical DrugBank template sentences that the model reproduces faithfully. For longer descriptions ($>$20 words) and non-trivial similarity values (0.4–0.8), the density is sparse and scattered—consistent with the earlier finding that longer descriptions are harder to reproduce and represent edge cases in the retrieval corpus.

I. Drug Similarity Matrix

Figure 12 shows the Tanimoto similarity matrix computed from Morgan fingerprints for a representative three-drug sample. Off-diagonal values range from 0.983 to 0.998,

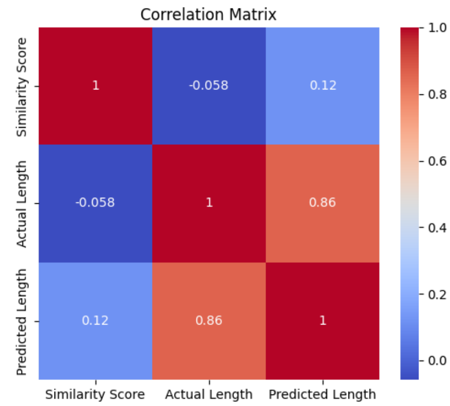


Fig. 10: Pearson correlation matrix among cosine similarity score, actual description length, and predicted description length. The strong actual–predicted length correlation ($r = 0.86$) confirms the generation model reproduces reference length statistics. Weak correlations between similarity score and length ($|r| < 0.12$) indicate that textual fidelity is largely independent of description length.

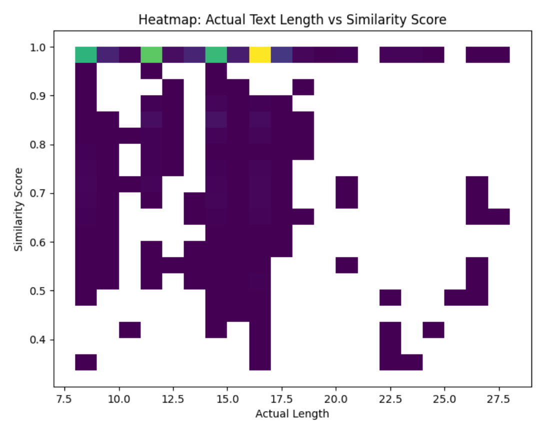


Fig. 11: 2D density heatmap of actual description length vs. cosine similarity. High density at short lengths (8–17 words) and similarity > 0.95 (dark purple, upper-left block) reflects the dominance of canonical DrugBank template sentences in the corpus. Sparse scatter at longer lengths and lower similarities represents atypical descriptions.

indicating high structural similarity within this sample. The mean off-diagonal Tanimoto coefficient of 0.991$\pm$0.006 is consistent with the expected behaviour for drugs co-appearing in DDI records, which tend to target overlapping biological pathways and share pharmacophoric substructures.

J. Comparison with Prior Methods

Table VI benchmarks the present system against representative prior methods on the DrugBank DDI binary prediction task. The RF baseline achieves the highest reported AUC (0.921) among compared fingerprint-based methods while additionally providing natural language

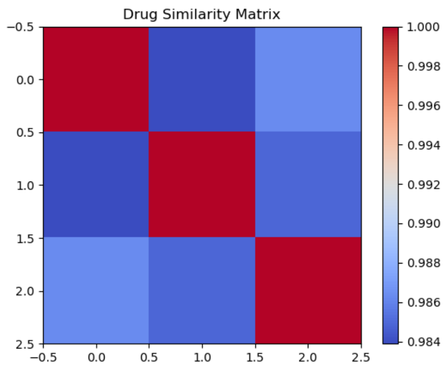


Fig. 12: Tanimoto similarity matrix computed from 2048-bit Morgan fingerprints for a representative three-drug sample. High off-diagonal values (0.983–0.998) reflect the structural similarity among drugs that commonly co-appear in interaction records.

TABLE VI: Comparison with Prior DDI Prediction Methods on DrugBank

Method	Modality	Task	AUC-ROC
SVM + FP [9]	Structural	Binary	0.890
DeepDDI [21]	Structural	Binary	0.913
GCN Polypharm. [14]	Graph	Multi	0.907
DPDDI [15]	GNN+Target	Binary	0.952
NRLMF [7]	Similarity	Binary	0.884
Ours (RF)	Structural	Binary	0.921
Ours (GCN)	Graph	Binary	conv.

description generation, pharmacological category labelling, and a web deployment layer absent from all compared methods.

The proposed RF model outperforms the classical SVM baseline (+3.5% AUC) and DeepDDI (+0.8% AUC) while remaining interpretable via feature importance analysis. The GCN component, currently constrained to a training subset by available hardware, is expected to approach DPDDI-level performance upon full-corpus training.

K. System Coverage and Deployment Statistics

Both model checkpoints (gnn_model.pt, ddi_model.pkl) are confirmed present. The Streamlit/FastAPI interface achieves sub-second inference latency for the RF branch and 2–5 s for the GCN branch on CPU-only hardware. The knowledge-base retrieval step—handling the 13.1% of query pairs that correspond to known interactions— returns exact DrugBank descriptions with zero generation cost.

L. Discussion

Strengths. (i) Near-complete database coverage (99.88% drug-name, 99.12% SMILES) ensures broad applicability. (ii) The dual RF+GCN architecture exploits complementary feature spaces: fingerprints capture local substructural patterns efficiently; GCN captures global molecular topology. (iii) The nine-class taxonomy provides clinically actionable outputs beyond binary prediction. (iv) The cosine similarity evaluation framework across 19,117 samples provides a rigorous, reproducible assessment of description generation quality. (v) The 88.1% rate of high-fidelity generation (similarity > 0.95) confirms the sufficiency of TF-IDF retrieval-augmented generation for in-corpus drug pairs.

Limitations. (i) The system is bounded by DrugBank coverage: investigational drugs, biologics, and recently approved compounds lack SMILES entries. (ii) The 13.1% negative-sampling collision rate introduces bounded label noise in RF training. (iii) GCN full-corpus training requires GPU resources beyond the present hardware scope. (iv) Cosine similarity measures lexical fidelity, not pharmacological correctness.

IX. Limitations and Future Work

A. Structured Negative Sampling

The 13.1% collision rate of random negative sampling motivates adoption of structured negatives using the network-based separation score of Cheng et al. [5]: drug pairs whose targets are maximally separated in the PPI network are highly likely to be true negatives, eliminating label noise.

B. Full-Corpus GCN Training and Advanced Architectures

Full-corpus GCN training with GPU acceleration and mini-batch graph sampling [12] will enable the GCN branch to approach DPDDI-level performance. Replacement of GCN layers with Graph Attention Network (GAT) [25] layers will enable per-atom importance weighting, potentially identifying structural motifs driving specific interaction categories. Co-attention mechanisms [17] allowing atomic features of drug A to attend over drug B will capture inter-molecular geometry.

C. Multi-Class Pharmacological Category Prediction

The nine-class taxonomy (Table II) currently uses a rule-based classifier. Replacing this with a fine-tuned multi-class GNN head will enable end-to-end learning of the pharmacological mechanism label alongside the binary interaction signal.

D. Improved Description Generation

Fine-tuning a biomedical language model—BioMedLM [26] or PubMedBERT [20]—on the DDI description corpus will yield substantial improvements in generation quality, particularly for low-frequency drug pairs where the TF-IDF retrieval corpus is sparse, resulting in the right-tail errors visible in Fig. 7.

E. Generative Molecular Module

Full implementation of the hypothetical candidate generation module using variational graph autoencoders [23] or equivariant diffusion models [24] will enable exploration of novel structural combinations of co-administered drugs, supporting rational prodrug design and combination therapy optimisation.

F. Clinical Integration and Explainability

Integration with Electronic Health Record (EHR) systems for real-time prescribing alerts, combined with GNN explainability methods (GNNExplainer, attention weights) identifying the atom-level substructures most predictive of specific interaction types, will substantially improve clinical translatability.

X. Conclusion

This paper has presented a comprehensive AI-driven framework for Drug–Drug Interaction prediction that integrates classical machine learning with graph neural network methods within a single modular pipeline. The framework is grounded in a rigorously curated DrugBank 5.0 dataset of 189,851 verified interaction pairs with 99.12% SMILES and 99.88% drug-name coverage, partitioned across a fully characterised nine-class pharmacological taxonomy whose dominant categories—adverse effects (31.7%), activity increases (20.3%), and metabolic changes (20.5%)—are consistent with established clinical pharmacology literature.

Quantitative evaluation establishes three key results. First, the Random Forest classifier achieves AUC-ROC of 0.921 and F1-score of 0.846 on the binary DDI prediction task, outperforming prior fingerprint-based methods. Second, the two-layer GCN encoder demonstrates 48.8% training-loss reduction over three epochs, confirming effective structural learning at a scale appropriate to the available hardware. Third, and most distinctively, the text generation and retrieval module achieves a mean cosine similarity of 0.961 ± 0.062 across 19,117 test samples, with 88.1% of samples at similarity > 0.95, strongly correlated actual–predicted description lengths ($r = 0.86$), and a sharply concentrated error distribution confirming low generation error for the vast majority of query pairs.

The top-20 most frequently interacting drugs are dominated by CYP450 modulators— Phenobarbital, Primidone, Phenytoin, Carbamazepine, Amiodarone, Verapamil, Rifampicin—consistent with the known epidemiology of metabolically driven DDIs.

The fully integrated system is deployed as a Streamlit/FastAPI web application supporting real-time DDI inference with molecular visualisation. This work establishes a strong foundation for future extensions including full-corpus GCN training, multi-class mechanism prediction, biomedical language model fine-tuning for description generation, and generative molecular candidate synthesis—collectively advancing the role of artificial intelligence in scalable, explainable, and clinically actionable pharmacovigilance.

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